

CRISTOFF STAN

Senior Software Engineer

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PROFESSIONAL SUMMARY

Innovative and performance-driven Senior Software Developer with over 9 years of hands-on experience designing, developing, and maintaining software across broadcasting, media, and enterprise domains. Specialized in C++, .NET, and C# development with solid experience in multimedia processing using frameworks like FFMPEG and GStreamer. Adept in building scalable APIs, working in cross-platform environments (Windows/Linux), and implementing CI/CD pipelines. Passionate about developing high-performance applications with robust architecture and clean code practices.

TECHNICAL SKILLS

- **Languages:** C++, C# (.NET), TypeScript, JavaScript, Python
- **Frameworks:** .NET 6/7, ASP.NET Core, GStreamer, FFMPEG, Node.js
- **Cloud & DevOps:** Docker, GitHub Actions, Jenkins, Kubernetes, CI/CD, Terraform
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis
- **Tools:** Git, Jira, Visual Studio, VSCode, Fiddler, Postman
- **Concepts:** OOP, SOLID, TDD, RESTful APIs, Microservices, Design Patterns, A/B Testing
- **Operating Systems:** Windows, Linux (Ubuntu, CentOS)

WORK EXPERIENCE

Senior Software Engineer

Sensidev

September 2021 – December 2024

Timișoara, Romania (Remote)

- Designed and developed microservice-based solutions using C++ and .NET Core for media asset management and live data processing tools used by broadcasting clients.
- Integrated FFMPEG and GStreamer pipelines to enable real-time media transformation, preview, and broadcast capabilities across live video feeds.
- Created RESTful APIs to interface video metadata, playback status, and device telemetry with a web-based control dashboard using Angular.
- Led migration of legacy monolithic systems into containerized microservices deployed with Docker and orchestrated via Kubernetes on a hybrid Windows/Linux cluster.
- Implemented asynchronous job processing using RabbitMQ and optimized inter-service communication through gRPC in high-throughput scenarios.
- Built automated CI/CD workflows with GitHub Actions for test-driven deployments, reducing release times by 40%.

- Mentored 4 junior developers and conducted regular code reviews, fostering clean coding practices and DevOps culture.
- Collaborated closely with designers, QA, and product managers to ensure delivery of user-centric, accessible features in visual storytelling platforms.

Software Engineer

Syscom Digital

October 2019 – August 2021

Bucharest, Romania (Remote)

- Developed and maintained enterprise-level broadcast automation tools using C++ and .NET Framework for media scheduling and campaign management.
- Implemented system-level integrations with digital signage platforms and third-party streaming services using HTTP/RTSP and socket-based protocols.
- Applied multithreading and memory optimization techniques in C++ to improve performance of media playback modules under load.
- Introduced Infrastructure-as-Code using Terraform to provision test environments, supporting faster iterations and release confidence.
- Participated in codebase modernization efforts, adopting C++17 features and unit testing frameworks to raise maintainability and test coverage.
- Integrated secure authentication mechanisms (OAuth2, JWT) into platform APIs, aligning with corporate infosec standards.
- Researched and embedded FFmpeg encoding configurations for adaptive bitrate streaming to meet client device requirements.
- Enhanced system observability by integrating Prometheus-based monitoring and structured logging across distributed modules.

Software Developer

Soft Build

June 2016 – July 2019

Timișoara, Romania (Remote)

- Built .NET-based automation software used in television broadcast control centers to manage transmission workflows and playlist generation.
- Led the implementation of cross-platform file management utilities in C++ that supported NTFS and ext4, improving asset ingestion performance.
- Applied the MVC pattern in building web-based dashboards for visualization of media asset statuses, logs, and alerts.
- Optimized SQL queries and indexing for PostgreSQL to reduce load times on real-time data dashboards by 55%.
- Collaborated with media engineers to debug and resolve codec compatibility issues across multiple ingest systems.
- Introduced static code analysis tools into build pipelines, reducing production incidents caused by regressions.
- Worked with legacy C# WinForms applications and migrated them into WPF with modern UI/UX enhancements.
- Wrote comprehensive documentation for the integration and deployment of video processing services across client sites.

Software Developer
Decima Digital
November 2015 – April 2016
Tallinn, Estonia

- Contributed to the development of internal broadcasting toolkits with C++ modules for content editing and timing synchronization.
- Implemented support for multiple audio track encoding in real-time capture software using FFmpeg.
- Debugged cross-platform inconsistencies during compilation of shared C++ libraries and implemented CMake standardization.
- Collaborated in developing a plugin architecture enabling media processing extensions, increasing code modularity and scalability.
- Wrote shell scripts to automate packaging and deployment of builds on internal Ubuntu and CentOS servers.
- Utilized Git for version control, managing branches across testing, development, and staging environments.
- Participated in SCRUM ceremonies and contributed to velocity improvements via better sprint planning.
- Assisted QA in writing unit and regression tests, achieving >90% test coverage in legacy modules.

Software Developer Intern
Borna Technology
September 2014 – August 2015
Tallinn, Estonia (On-Site)

- Developed internal broadcasting workflow automation tools in C# and contributed to supporting scripts written in Python and Bash.
- Assisted in integrating third-party APIs and broadcast systems using SOAP and RESTful endpoints.
- Built internal CLI tools in C++ for batch media conversion tasks, handling thousands of files per execution.
- Conducted testing and validation of media playback behavior across Windows and Linux player platforms.
- Helped maintain documentation for API usage, testing guides, and CI scripts.
- Actively participated in code reviews and weekly developer syncs to learn modern software practices.
- Created dashboards using .NET WinForms to display system performance and real-time broadcast indicators.
- Supported the DevOps team in debugging Jenkins-based CI/CD failures and environment inconsistencies.

EDUCATION

Tallinn University

Bachelor's Degree in Computer Science,
2010 – 2014

CERTIFICATIONS

- ❖ Microsoft Certified: Azure Developer Associate
 - ❖ C++ Institute CPA: C++ Certified Associate Programmer
 - ❖ Certified Kubernetes Application Developer (CKAD)
 - ❖ FFmpeg Advanced Media Processing – Udemy
 - ❖ GitLab CI/CD for Developers – LinkedIn Learning
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PROJECTS

Live Media Control Suite

- Architected a C++ and .NET hybrid application to handle live channel switching and content previewing in news studios, integrated with GStreamer for real-time feed parsing and adaptive layouts.

Cloud Broadcast Orchestrator

- Built a scalable microservices-based control panel using .NET and Docker that orchestrates encoding jobs via FFMPEG across Kubernetes clusters.

Video Sync Timeline Debugger

- Developed a debugging interface for real-time timeline sync across multiple live input feeds using C++20 coroutines and observable logging channels.